



Curriculum Vitae of

Emmanuel Deogratias

BED & MSc (University of Dar es Salaam, Tanzania)

EDD (University of Alberta, Canada)

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1.0 Contacts

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2.0 Biodata

Year of Birth Date: 25/12/1982

Place of Birth: Ilemela, Mwanza

Place of Domicile: Nansio Ukerewe

Tribe: Jita

Sex: Male

Nationality: Tanzanian

Marital status: Married

3.0 Areas of Specialization

Mathematics Education

4.0 Academic Credentials

2014 –2020 Doctor of Education in Mathematics Education., Department of Secondary Education, University of Alberta.

Dissertation title: *Exploring the Implementation of Concept-Rich Instruction with University Mathematics Pre-Service Teachers. A Tanzanian Case.*

2009 – 2011 MSc Mathematical Modelling., Department of Mathematics, University of Dar es Salaam.

Dissertation title: *Methods for Pricing and Hedging Plain Vanilla Barrier options.*

- 2005 – 2008 B. ED (Majoring mathematics and Education)., Faculty of Education, University of Dar es Salaam.
- 2002 – 2004 Advanced Secondary School Education., Musoma High School, Tanzania. Majoring: *Mathematics, Chemistry and Physics.*
- 1998 – 2001 Ordinary Secondary School Education., Bukongo Secondary School in Ukerewe District, Tanzania.
- 1991 – 1997 Primary School Education., Mahande Primary School in Ukerewe District, Tanzania.

5.0 Skills

- 5.1. Proficiency in using qualitative data analysis tools (e.g., NVivo) and statistical analysis software (e.g., SPSS). NVivo was used while analysing data collected from pilot and main studies for my doctoral research work. SPSS is being used while teaching and supervising my undergraduate and postgraduate students during analysing quantitative data.
- 5.2. Proficiency in English and Kiswahili languages.
- 5.3. Experiences in conducting both qualitative and quantitative research. I employed qualitative research approach during my doctoral research work while quantitative research was used during conducting my master degree research work.
- 5.4. Skills in engaging stakeholders and local communities in education through participatory techniques. Community engagement in education in Morogoro, Iringa and Dodoma was conducted through capacity building project funded by Global Affairs Canada. The project was a coalition of Brock University, University of Alberta and University of Dodoma.
- 5.5. Strong analytical and report writing skills.

6.0 Selected Working Experience/Employment

- 2025 – Member of College Committee for Research and Publications, College of Natural and Applied Sciences, Sokoine University of Agriculture, Tanzania.
- 2025 – Coordinator for Outreach program, Department of Mathematics and Statistics, Sokoine University of Agriculture, Tanzania.
- 2023 – Senior Lecturer, Department of Mathematics and Statistics, Sokoine University of Agriculture, Tanzania.
- 2023 – Editorial Review Board Member, International Journal of Teacher Education and Professional Development, IGI Global Publisher, United States.
- 2022 –2023 Lecturer, Department of Mathematics and Statistics, Sokoine University of Agriculture, Tanzania.
- 2022 – Member of College Academic Committee, College of Natural and Applied Sciences, Sokoine University of Agriculture, Tanzania.
- 2021—2022 Best worker at the university level, Department of Mathematics and Statistics

- University of Dodoma, Tanzania.
- 2021—2022 Online Global Social Justice in Education Course Leader, College of Education, Purdue University, United States of America.
- 2020—2022 Coordinator for Research and Publications, Department of Mathematics and Statistics, University of Dodoma, Tanzania.
- 2020—2022 Lecturer, Department of Mathematics and Statistics, University of Dodoma.
- 2014 – 2018 Graduate Research Assistant, Department of Secondary Education, University of Alberta, Canada.
- 2013 – 2014 Acting Head of Department, Mathematics Department, University of Dodoma, Tanzania.
- 2012 – 2014 Undergraduate Coordinator, School of Mathematical Sciences, University of Dodoma, Tanzania.
- 2011 – 2020 Assistant Lecturer, Department of Mathematics, University of Dodoma
- 2009 – 2011 Tutorial Assistant, Department of Mathematics, University of Dodoma.
- 2008 – 2009 Tutor, Dakawa Diploma Teachers College, Morogoro, Tanzania.
- 2008 – 2009 Registrar, Dakawa Diploma Teachers College, Morogoro, Tanzania.
- 2004 – 2005 Mathematics teacher, Murutunguru Secondary School in Ukerewe District, Tanzania.

7.0 Selected Publications

- 7.1 Deogratias, E. (2026). The role of complex numbers in interdisciplinary mathematics teaching in Tanzanian secondary schools. *Journal of Mathematics and Science Teacher*, 6(2), em107. <https://doi.org/10.29333/mathsciteacher/18344>
- 7.2 Deogratias, E., Gervas, I., Kazuzuru, B., & Bukuku, S. (2026). Budgeting Practices of Secondary Schools in Allocating Operational Funds for Instructional and Support Services in Mbeya District, Tanzania. *International Journal of Curriculum and Instruction* 18(2), 44–56.
- 7.3 Deogratias, E., Mathias, G., & Lupeja, T. (2026). Mathematics Teachers' Competencies in Designing Lessons that Foster 21st-Century Skills Acquisition Among Secondary School Students in Kwimba District, Tanzania. *International Journal of Curriculum and Instruction* 18(1), 1–24.
- 7.4 Deogratias, E., Mwiyaenzi, J., & Karugila, G. (2025). Secondary school students' perceptions of physics and interdisciplinary course combinations in Tanzania. *Union: Jurnal Ilmiah Pendidikan Matematika*, 13(4), 1050-1065.
- 7.5 Deogratias, E., Mathias, G., & Lupeja, T. (2025). Exploring mathematics teachers' instructional and formative assessment practices for fostering 21st-century skills. *Union: Jurnal Ilmiah Pendidikan Matematika*, 13(3), 819-838.
- 7.6 Deogratias, E., Hezron, M., & Lupeja, T. (2025). Examining Ordinary Level Secondary Students' Connections Between Classroom-Learned Mathematics with Their Real-Life Experiences. *International Journal of Research in Mathematics Education*, 3(2), 191-202.
- 7.7 Deogratias, E., Sarakikya, E., Kira, E., Lupeja, T., & Karugila, G. (2025). Enhancing Conceptual Understanding in Mathematics: The Effectiveness of the Square Method for Computing Areas of Irregular Shapes Among Secondary School Students in Tanzania. *Education XXI Journal*, 28(4). doi:10.5944/vol28n4a309
- 7.8 Deogratias, E., Neterindwa, A., Mkwizu, M., Iddi, A., Karugila, G., Matofali, A., & Nyerere, N. (2025). Enhancing Understanding of Fraction Division using the Plain Paper Method in Tanzanian Secondary Schools. *Liberte Journal*, 13(10), 247-273.
- 7.9 Deogratias, E., Mkwizu, M., & Iddi, A. (2025). Enhancing understanding of set representation through Venn diagrams: The effectiveness of integrating mathematics

student teachers' lived experiences. *Oeil Research Journal*, 23(10), 137-162.

- 7.10 Deogratias, E., & Iddi, A. (2025). Investigation on the factors leading to negative attitudes towards mathematics among secondary school students in Tanzania. *Union: Journal of Pendidikan Matematika*, 13(3), 761-788
- 7.11 Deogratias, E., & Matumba, E. (2025). Investigation on the impact of learning mathematics using mobile phones at higher institutions in a Tanzanian context. *Union: Jurnal Ilmiah Pendidikan Matematika*, 13(2), 351-366.

- 7.12 Deogratias, E., & Karugila, G. (2025). Estimation of the Population Size in Tanzania from 1967 to 2112 Using Logistic and Exponential Models: A Comparative Analysis. *Journal of Humanities, Arts and Social Science*, 9(6), 1201-1218.
- 7.13 Deogratias, E., & Paulo, H. (2025). Investigating the Factors Contributing to Low Mathematics Performance in Tanzanian Ordinary Secondary Schools. *International Journal of Research in Mathematics Education*, 3(1), 91-110.
- 7.14 Deogratias, E., Mkwizu, M., Msangi, N., Matofali, A., & Karugila, G. (2025). A detailed descriptive study on the teaching of odes to a group of preservice teachers using computer assisted instruction. *Proceedings of the 8th World Conference on Future of Education*. Cambridge University: United Kingdom.
- 7.15 Deogratias, E., & Mrope, F. M. (2025). Using a crossing method as alternative approach for teaching, learning and solving quadratic equations. *The Educational Review, USA*, 9(4), 396-408.
- 7.16 Bukuku, S., Deogratias, E., Kazuzuru, B., & Gervas, I. (2025). Challenges facing management of operational funds allocated for instructional and support needs of secondary schools in Mbeya District, Tanzania. *International Journal of Curriculum and Instruction*, 17(2), 627-642.
- 7.17 Deogratias, E., Ngonyani, C., & Gadala, Y. (2025). Enhancing mathematical understanding through real objects: Insights from Tanzanian classrooms. *Union: Jurnal Ilmiah Pendidikan Matematika*, 13(1), 1-14. <https://doi.org/10.30738/union.v13i1.18733>
- 7.18 Deogratias, E., & Karugila, G. (2024). Overcoming students' misuse of linearity in a non-linearity situation in schools. *The Educational Review, USA*, 8(12), 1418-1426.
- 7.19 Mrope, F. M., Kira, S. E., & Deogratias, E. (2024). The influence of peer groups on academic performance in University level mathematics courses in: A case study of Morogoro municipality. *International Journal of Research in Mathematics Education*, 2(2), 149-162.
- 7.20 Mrope, F. M., & Deogratias, E. (2024). Analysis of students' difficulties in solving probability questions: The case of advanced certificate of secondary education examination in Tanzania. *Union: Jurnal Ilmiah Pendidikan Matematika*, 12(3), 494-506. <https://doi.org/10.30738/union.v12i3.17730>
- 7.21 Langoi, S., & Deogratias, E. (2024). Investigation of teachers' competency on using ICT for teaching and learning mathematics and science subjects in Tanzanian secondary schools. *International Online Journal of Education and Teaching*, 11(3), 619-632.
- 7.22 Sapkota, B., Leo, X., Sapkota, M., Akarsu, M., Deogratias, E., Fauber, D., Mbewe, R., Mumba, F., Panthi, R. K., Newton, J., & Phillion, J. (2024). Exploring international educators' learning about local and social justice in a virtual community of practice. In A. Slapac & C. Huertas-Abril (Eds.), *Encouraging transnational learning through telecollaboration in global teacher education* (pp. 45-78). IGI Global. <http://dx.doi.org/10.4018/978-1-6684-7813-4>
- 7.23 Deogratias, E., Komanya, K., & Nicolaous, G. (2024). Investigating the Effect in Teaching and Learning Mathematics while Arranging Students in Classes According to Their Performance in Tanzanian Secondary Schools: Teaching and Learning Mathematics while Arranging Students in Classes. *International Journal of Curriculum and Instruction*, 16(3), 627-642.
- 7.24 Hezron, S., Deogratias, E., & Lupeja, T. (2024). Exploring the Extent at Which Mathematics Teachers Use Real-Life Examples/Aids/Situations in Their Daily Mathematics Teaching and Learning Environment: Real-Life Examples/Aids/Situations in Daily Mathematics for Teaching and Learning Environment. *International Journal of Curriculum and Instruction*, 16(3), 669-680.

- 7.25 Deogratias, E., Rao, G. S., & Alphonse, F. (2024). Evaluating students' performance in mathematics and other subjects using fixed and individual control chart grading methods. *Journal of Mathematics and Science Teacher*, 4(3).
- 7.26 Kornelio, S., Balan, R., & Deogratias, E. (2024). Forecasting students' enrolment in Tanzania government primary schools from 2021 to 2035 using ARIMA model. *International Journal of Curriculum & Instruction*, 16(1), 162-174.
- 7.27 Deogratias, E., & Mrope, F. (2024). Using crossing method for teaching, learning and solving systems of linear equations for two unknowns that yield no solution in Tanzanian secondary schools. *International Journal of Curriculum and Instruction*, 16(2), 434-440.
- 7.28 Idd, A., Mbeho, N., & Deogratias, E. (2023). Reflective questions on relation on a set: Learning from undergraduate students taking computer networks and information security engineering program. The 5th International Conference on Innovative Research in Education (pp. 38-51). Diamond Scientific Publishing. doi: <http://dx.doi.org/10.33422/5th.ireconf.2022.11.450>
- 7.29 Deogratias, E. (2023). Enhancing Pre-Service Teachers' Understanding of A Mathematical Concept Through Practises: A Constructivist Perspective. *Proceedings of the 5th International Academic Conference on Education*. Denmark: Copenhagen. doi: <https://www.doi.org/10.33422/5th.iaceducation.2022.07.13>
- 7.30 Alphonse, F., Rao, GS., & Deogratias, E. (2022). Students' performance evaluation for secondary school in Mbulu district using mean and dispersion charts grade and fixed grading method. *Biometrics & Biostatistics International Journal*, 11(4):152–157. DOI: 10.15406/bbij.2022.11.00370
- 7.31 Deogratias, E. (2022). Cognitive development of children's understanding of counting numbers using a rope. *International Journal of Curriculum and Instruction*, 14(3), 1800-1812.
- 7.32 Deogratias, E. (2022). Using graph coloring for effective timetable scheduling at ordinary secondary level. *International Journal of Curriculum and Instruction*, 14(2), 1166-1188.
- 7.33 Deogratias, E. (2022). Using a crossing method as an alternative approach for teaching systems of linear equations in secondary schools. *International Online Journal of Education and Teaching (IOJET)*, 9(2), 1022-1031.
- 7.34 Deogratias, E. (2022). The importance of using real objects for teaching and learning a mathematical concept with pre-service teachers of mathematics. *International Journal of Curriculum and Instruction*, 14(1), 24-36.
- 7.35 Deogratias, E. (2022). Developing student-teachers' understanding of metric and topological spaces using reflective questions. *International Online Journal of Education and Teaching*, 9(1), 1-10.
- 7.36 Deogratias, E. (2021). Developing student-teachers' understanding of geometrical objects/figures using a bicycle rubber tube. *World Journal of Educational Research*, 8(5), 1-9.
- 7.37 Deogratias, E. (2020). Learning by doing: Informed by university mathematics students' participation in field practical training. *International Journal of Science and Research*, 9(11), 1404-1405.
- 7.38 Deogratias, E. (2020). *Exploring the implementation of concept-rich instruction with university mathematics pre-service teachers: A Tanzanian Case* (Doctoral dissertation, University of Alberta, Edmonton, Canada).
- 7.39 Deogratias, E. (2020). Forms of knowledge and knowing in mathematics education: informed by concept study based on complexity theory. *International Journal of Curriculum and Instruction*, 12(2), 191–208.
- 7.40 Deogratias, E. (2019). The efficacy of concept-rich instruction with university pre- service teachers in a Tanzanian context using Vygotskian perspective. *World Journal of Educational Research*, 6(3), 373—385.

- 7.41 Deogratias, E. (2018). The possible ways of practicing complexity theory through concept study in mathematics class. *International Journal of Curriculum and Instruction*, 10(2), 142—151.
- 7.42 Deogratias, E. (2017). Informing local materials used for concept-rich instruction practice in mathematics class with pre-service teachers in Tanzanian context. In J. Holm., S. Mathieu., & S. Oesterle (Eds.), *Proceedings of the 2017 Annual Meeting of the Canadian Mathematics Education Study Group* (pp. 271—272). Montreal: McGill University. Retrieved from <https://files.eric.ed.gov/fulltext/ED589990.pdf>
- 7.43 Deogratias, E. (2013). *Methods for pricing and hedging plain vanilla barrier options* (Master Dissertation, University of Dar es Salaam, Tanzania).

7.0 Selected International Conferences

- 7.1 Deogratias, E., Mkwizu, M., Msangi, N., Matofali, A., & Karugila, G. (2024). A detailed descriptive study on the teaching of odes to a group of preservice teachers using computer assisted instruction. *Proceedings of the 8th World Conference on Future of Education in Cambridge. United Kingdom*.
- 7.2 Deogratias, E., Mbeho, N., & Iddi, A. (2022). Reflective questions on relation on a set: learning from undergraduate students taking computer networks and information security engineering program. The 5th International Conference on Innovative Research in Education in Amsterdam. Netherlands.
- 7.3 Deogratias, E. (2022). Enhancing pre-service teachers' understanding of a mathematical concept through practices: A constructivist perspective. 5th International Academic Conference on Education in Copenhagen. Denmark.
- 7.4 Deogratias, E. (2016). Informing a concept study practice in mathematics class from North America to a Tanzanian context: ICME-13 Conference in Hamburg. Germany.

8.0 Presentations

- 8.1. Deogratias, E., Mkwizu, M., Msangi, N., Matofali, A., & Karugila, G. (2024). *A detailed descriptive study on the teaching of odes to a group of preservice teachers using computer assisted instruction*. Paper presented on the 8th World Conference on Future of Education in Cambridge University. United Kingdom.
- 8.2. Deogratias, E., Mbeho, N., & Iddi, A. (2022). *Reflective questions on relation on a set: learning from undergraduate students taking computer networks and information security engineering program*. Paper presented on the 5th International Conference on Innovative Research in Education in Amsterdam. Netherlands.
- 8.3. Deogratias, E. (2022, July). *Enhancing pre-service teachers' understanding of a mathematical concept through practices: A constructivist perspective*. Paper presented for the 5th International Academic Conference on Education in Copenhagen, Denmark.
- 8.4. Deogratias, E. (2018, March). *University pre-service teachers expressions of their understanding of pi through concept-rich instruction: A Tanzanian case*. Poster presented for Graduate Student Showcase at the University of Alberta, Edmonton, Canada.
- 8.5. Deogratias, E. (2018, January). *Activities based on concept-rich instruction: A possible way to improve university pre-service teachers' understanding of the mathematical concepts for teaching*. Poster presented for MathEd Forum, Fields Institute for Research in Mathematical Sciences, Canada.
- 8.6. Deogratias, E. (2017, June). *Informing local materials used for concept-rich instruction practice in mathematics class with pre-service teachers in Tanzanian context*. Poster

presented for the 2017 Annual Meeting of the Canadian Mathematics Education Study Group, McGills University, Canada.

- 8.7. Deogratias, E. (2016, July). *Informing a concept study practice in mathematics class from North America to a Tanzanian context*. Poster presented for ICME-13 Conference in Hamburg, Germany.
- 8.8. Wang, X., & Deogratias, E. (2015, February). *Evaluating knowledge level and cognitive demands of task in grade eight mathematics textbooks*. Poster presented at Phi Delta Kappa (PDK) International graduate research, University of Alberta, Canada.

9.0 Selected Workshops/Short Courses/Seminars Participated

- 9.1. Certificate of participation in the Webinar on how to publish your research with Routledge by the Association of African Universities and Taylor & Francis Group, held on June 15th 2023.
- 9.2. Innovators and entrepreneurs' mentorship program participation, held in September 2022, at Kings Hotel in Morogoro by Innoversity Project. The project is funded by France Embassy under Sahara Venture.
- 9.3. Workshop facilitation on professional development for university instructors: Improving the quality of teaching and learning approaches under Transforming Employability for Social Change in East Africa (TESCEA) project, held in November 15-16, 2021 at the College of Natural and Mathematical Sciences in the University of Dodoma.
- 9.4. Record of completion of online six week training course for becoming a facilitator in East Africa for transformative teaching and learning in Universities under TESCEA project, completed between October 26, 2021—December 14, 2021. The course consisted of the following modules:(1) Developing your teaching philosophy, (2) What is a quality learning design? (3) The Learning Designer tool, (4) ILOs and TLAs - Good practice, and (5) Aligning ILOs, TLAs and learning assessment. The course concluded with the submission and peer assessment of a learning design.
- 9.5. Workshop participation on developing entrepreneurial teachers training for transformation under Sustainable Development Goals (SDG) Agents focusing on entrepreneurial thinking, sustainable development and ICT skills, held online from October 14, 2021 to November 18, 2021. SDG Agents is the project of the University of Vechta in Germany in partnership with Nelson Mandela University in South Africa. Through this workshop, I developed a prototype/Idea Hub on School Entrepreneurship Shop, testing the idea at primary school and evaluating the idea if it is practical and sustainable.
- 9.6. Workshop facilitation on five-day course redesign agenda under TESCEA project focusing on concept mapping, learning outcomes, learning assessment, feedback, learning activities, learning approaches/strategies, learning design, gender responsiveness pedagogy, and lesson planning, held in September 21-25, 2021 at the College of Social Sciences in the University of Dodoma.
- 9.7. Team leader and participant on global social justice in (mathematics) education courses, held from November 2020 to December 2021 online under the collaboration of the College of Education at Purdue University in the United States of America.

- 9.8. Innovators and entrepreneurs' mentorship program participation, held from April to June 2021 at CIVE Innovation and Entrepreneurship Center in the University of Dodoma.
- 9.9. Workshop participation on research, publications and consultancy, held in May 2021 at the College of Natural and Mathematical Sciences in the University of Dodoma.
- 9.10. Workshop facilitation on enhancing pedagogical knowledge and skills for university instructors, held in April 2021 at the College of Natural and Mathematical Sciences in the University of Dodoma.
- 9.11. Workshop facilitation on professional development for university instructors: Construction of a test and university examination using Bloom's taxonomy, held in February 2021 at the College of Natural and Mathematical Sciences in the University of Dodoma.
- 9.12. An evening symposium showcasing collaborative research projects between Alberta Schools and University of Alberta, sponsored by CMASTE and supported by Alberta Education Research Partnerships program, held in November 7, 2019 at Glacier Room, Lister Center, University of Alberta. There were two presentations: Predictors of performance on the PATs by Lynn McGarvey and Okan Bulut (University of Alberta) and impact of the co-development of a mathematics framework by Jason Drent and Lissa Steele (Chinook's Edge School Division).
- 9.13. CMASTE monthly seminar series participation about mathematics, science and technology education from September 2014—November 2019, held in Education South, Faculty of Education, University of Alberta.
- 9.14. Workshop participation focused on reviewing the developed 5 draft policy briefs related to ongoing professional learning, gender, empowering schools, parental engagement, and teacher resource centres, held in November 2017 at the University of Dodoma under the project for Capacity Development for Mathematics Teaching in Rural and Remote Communities in Tanzania conducted by University of Dodoma, Tanzania under collaborations with University of Alberta (Canada) and Brock University, Canada.
- 9.15. Seminar series participation about "Getting to Know the Canadian and Alberta Educational Contexts: A Seminar Series for International & Newly-Immigrated Graduate Students" from September – November, 2015 held in the Department of Secondary Education, University of Alberta.
- 9.16. Teacher as a resource short course held from 10-21, February, 2014 at VETA Dodoma under the project for Capacity Development for Mathematics Teaching in Rural and Remote Communities in Tanzania conducted by University of Dodoma, Tanzania under collaborations with University of Alberta (Canada) and Brock University, Canada.
- 9.17. Record of completion of graduate ethic training course (5 hours) in the faculty of graduate studies and research, University of Alberta, completed between September 2013—August 2014.
- 9.18. University Quality Assurance Framework 8-11 workshop, held in April, 2014 at University of Dodoma, CIVE Computer LAB2.
- 9.19. Professional teaching methodologies at universities workshop held at the University of Dodoma in February 2012.

10.0 Selected Projects/Consultancies

- 10.1. Deogratias and colleagues (2022). Editors Advanced Mathematics for Secondary Schools Student's Book Form Six, Tanzania Institute of Education.
- 10.2. Deogratias and colleagues (2021). Editors Basic Mathematics for Secondary Schools Student's Book Form Four, Tanzania Institute of Education.
- 10.3. Deogratias and colleagues. (2017). Reviewing the developed 5 draft policy briefs related to ongoing professional learning, gender, empowering schools, parental engagement, and teacher resource centres, *Hisabati ni Maisha Project*.

11.0 Ph.D. Courses Studied at the University of Alberta

- EDSE 501 Conference seminar: Mathematics for teaching.
- EDSE 503 Curriculum foundation.
- EDSE 504 Curriculum inquiry.
- EDSE 510 Research methods in secondary education.
- EDSE 539 Research issues in teaching and learning mathematics.
- EDSE 540 Secondary mathematics education: Examining tasks, curriculum and programs.
- EDSE 601 Advanced guided independent study in secondary education: Introduction to school improvement.
- EDSE 602 Advanced guided independent study in secondary education: Leadership in educational settings.
- EDSE 610 Advanced research seminar in secondary Education I.
- EDSE 620 Advanced research seminar in secondary Education II.
- EDPS 671 Issues in administration of post-secondary education.

12.0 Paper Reviews

- 12.1. Deogratias, E. (2025, July). European Journal of Science and Mathematics Education, Springer [Review of a paper *A systematic review of the role of visualization in secondary school mathematical problem solving*, by Author X].
- 12.2. Deogratias, E. (2023, September). International Journal of Teacher Education, IGI Global [Review of a paper *Teachers' attitudes towards the effectiveness of collaborative reflection supportive method: The impact of achievement goal and community identification*, by Author X].
- 12.3. Deogratias, E. (2023, September). International Journal of Teacher Education, IGI Global [Review of a paper *Teachers' attitudes towards the effectiveness of collaborative reflection supportive method: The impact of achievement goal and community identification*, by Author X].
- 12.4. Deogratias, E. (2023, July). European Journal of Science and Mathematics Education, Springer [Review of a paper *Assessing the added value of a history-based activity for students with low mathematics skills*, by Author X].
- 12.5. Deogratias, E. (2023, April). IGI Global Publisher [Review of a Book Chapter *Virtual Exchange in Teacher Training and the Foreign Language Classroom: Global Competence Skills and Development in an Interdisciplinary Pilot*, by Authors X].

- 12.6. Deogratias, E. (2023, April). International Online Journal of Education and Teaching (IOJETr) [Review of a paper *Developing Didactic Games to Improve the Mathematical Skills of Young Children*, by Authors X].
- 12.7. Deogratias, E. (2023, April). Journal of Mathematics and Science Teacher, Modestum [Review of a paper *Emerging Perspective of a Mathematical Teacher Educator on Connections Between Advanced and School Mathematics*, by Authors X].
- 12.8. Deogratias, E. (2022, November). International Electronic Journal of Mathematics Education, Modestum [Review of a paper *Teachers' Appraisal of Various Aspects of the Efficiency of 1 Homogeneous and Heterogeneous Learning Groups*, by Authors X].
- 12.9. Deogratias, E. (2022, September). Review of Educational Research, SAGE [Review of a paper *Lectures in Higher Education: A 22-Year Systematic Review*, by Authors X].
- 12.10. Deogratias, E. (2022, April). International Online Journal of Education and Teaching (IOJETr) [Review of a paper *A Comparison of Secondary School Students' Value Perceptions in Turkish*, by Authors X].
- 12.11. Deogratias, E. (2022, January). European Journal of Science and Mathematics Education, Springer [Review of a paper *Teacher education and didactics knowledge to teach statistics: A case study*, by Authors X].
- 12.12. Deogratias, E. (2021, September). European Journal of Science and Mathematics Education, Springer [Review of a paper *Mathematics teaching efficacy belief and attitude of pre-service 1 teachers and academic achievement*, by Authors X].
- 12.13. Deogratias, E. (2021, September). International Online Journal of Education and Teaching (IOJETr) [Review of a paper *An Investigation of Students' Quantitative Reasoning Through Modeling Process*, by Authors X].
- 12.14. Deogratias, E. (2021, August). European Journal of Science and Mathematics Education, Springer [Review of a paper *CHEMBOND3D e-Module Effectiveness in Enhancing Students' Knowledge of Chemical Bonding Concept and Visual-spatial Skills*, by Authors X].
- 12.15. Deogratias, E. (2021, August). International Online Journal of Education and Teaching (IOJETr) [Review of a paper *The created learning activities for employees to enhance quality of work of small businesses in Thailand*, by Authors X].
- 12.16. Deogratias, E. (2018, March). Graduate Student Showcase at the University of Alberta [Review of a paper *The recursive interaction of mathematics teachers in a Blog: Complexity inquiry* by X].
- 12.17. Deogratias, E. (2016, February). ICME-13 conference in July 24-31, 2016 in Humburg, Germany [Review of a paper *Student teacher's pedagogical competency in lesson study and open approach* by X].

13.0 Supervised Master and PhD Students

- 13.1. **Enezael N. Kyungai:** (PHD in Education Foundation and Management at Sokoine University of Agriculture) (On Progress)

- 13.2. **Wantrudis Claud Mkomange:** (PHD in Mathematics Education at Sokoine University of Agriculture) (On Progress)
- 13.3. **Mwema Robi Felix:** (PHD in Curriculum and Instruction at Sokoine University of Agriculture) (On Progress)
- 13.4. **George Mathias:** (Master's in mathematics education at Sokoine University of Agriculture) (Completed)
- 13.5. **Hezron Majura:** (Master's in mathematics education at Sokoine University of Agriculture) (Completed)
- 13.6. **Sophia Kornelio.** (2022). Forecasting Students' Enrolment in Tanzania Government Primary Schools Subjected to Classroom Infrastructure, Department of Mathematics and Statistics, University of Dodoma (Master Dissertation submitted).
- 13.7. **Filbert Alphonse.** (2022). Comparing Evaluation of Students' Performance in Secondary Schools in Mbulu District Using Statistical Quality Control and Fixed Grading Method, Department of Mathematics and Statistics, University of Dodoma (Master Dissertation submitted).

14.0 Supervised Undergraduate Research Projects at the University of Dodoma

- 2022 Salu Mabawan, Adras Lugee, Elisha Mwambashi, Steven Rukra, Juma Said Juma, Frank Cosmas and Israel Baseka, Statistical analysis on water demand, supply and consumption in Dodoma city, University of Dodoma.
- 2021 Marko Gervas, Mashala Charles, Hatibu Lidia and Libanda John, Using Graph Coloring for Effective Timetable Scheduling at Ordinary Secondary School, University of Dodoma.
- 2021 Yapemungu Samweli and Colleagues, Optimization of Water Use for Domestic's Sewage Systems Using Metric and Topological Spaces, University of Dodoma.
- 2014 Lucas Pastory, Ally Lushino, Omari Badili and Lebahati Kinisi, Mathematics teaching and learning resources in Tanzania secondary schools: Experiences from selected secondary schools of Dodoma municipality.
- 2014 Pius Mussa, Ashery Boniface, Alphonse Donath and Michael Simba, The influence of mathematics in understanding science subjects: A case study of Dodoma municipality.
- 2014 Kizith Metili, Albertho Mkinga, Joseph Suleman and Endrick Yohana, Correlation between mathematics and other subjects: A case study of Dodoma municipality.
- 2014 Humphrey Kalisa, Jeannette D'Mello, Rashoudy Bansagwile and Afisai Julius, The relationship between mathematics and culture (ethnomathematics) in a society. A case study of Dodoma municipality.
- 2013 James Othniel, Lemboy Kivuyo and Tungu Martinde, Why do Most Students in Tanzania Secondary Schools Dislike Mathematics? A Case study in Dodoma Municipal, University of Dodoma.
- 2013 Veronica Ligogo and Mchani Mdogo, Applications of Partial Differential Equations in a Vibrating Membrane, University of Dodoma.
- 2013 Lukindo Seif and Kilimo Thomas, Mathematics as a Challenging subject at Primary and Secondary Schools in Tanzania. A Case Study in Dodoma Municipal, University of Dodoma.

- 2012 Honory Nicodemas and John Andrea, Methods for Pricing European Options, University of Dodoma.
- 2012 Mghaya Novati and Ngogo Shahabu, Study on Methods for Pricing Barrier Options, University of Dodoma.
- 2012 Mwaipungu Patrick and Temba Aloyce, Types of Forces and their Applications,

University of Dodoma.

2012 Herman Kamuntu, Pricing American Options, University of Dodoma.

15.0 Undergraduate and Master Courses Taught at the University of Dodoma

MT 1111: Discrete mathematics
MT 1114: Business mathematics
MT 112: Linear algebra
MT 122: Ordinary differential equations
MT 214: Rigid body mechanics
MT 221: Complex analysis
MT 320: Mathematics project
MT 321: Basic functional analysis
MT 323: Tensor analysis
MT 613: Metric and topological spaces
MT 621: Functional analysis
MT 623: Number theory

16.0 Internal Examiner for Master and PhD Students

17.0 Volunteer Work While Pursuing Doctoral Studies at the University of Alberta, Canada

- 16.1. Church volunteering services, such as cleaning, LDS Church at Bearspaw Chapel, Edmonton, Canada.
- 16.2. Elevator operator Move in Day (1 spot) from 2:15 pm to 5:30 pm on Monday August 26, 2019, By Alberta Room, Lister Center, University of Alberta.
- 16.3. Community Liaisons (1 spot) from 8:00 am to 12:00 noon on Monday June 13, 2016, and (1 spot) from 12:00 noon to 4:00 pm both on Tuesday June 14 and Wednesday, June 15, 2016, By Alberta Room, Lister Center, University of Alberta.
- 16.4. What Makes Vegreville a Healthy Community? Community Forum held on February 11, 2016 at the Pomeroy Hotel in Vegreville, Alberta (A project funded by Alberta Health).